DS701 Quiz 2 Reference Sheet
Lectures 08–12 · Supervised learning and representation

08 · Decision Trees & Random Forests

- **Tree:** internal node = attribute test; leaf = class label. Classify by walking root to leaf. Grow greedily: pick the best split, recurse on children until a *stopping criterion* (pure node, `min_samples_split`, `max_depth`).
- **Impurity at node** t ($p_i(t)$ = fraction of class i, c classes; $0 \log_2 0 = 0$):

$$\text{Gini} = 1 - \sum_i p_i^2$$

$$\text{Entropy} = -\sum_i p_i \log_2 p_i$$

$$\text{Error} = 1 - \max_i p_i$$

- All are 0 for a pure node, maximal for uniform classes (binary: Gini $\frac{1}{2}$, entropy 1, error $\frac{1}{2}$).
- **Collective impurity of children** ($N(v_j)$ instances at child v_j, k children):

$$I(\text{children}) = \sum_{j=1}^{k} \frac{N(v_j)}{N} I(v_j)$$

- **Information gain:** $\Delta_{\text{info}} = I(\text{parent}) - I(\text{children})$; choose the split with the largest gain (smallest collective impurity).
- **Gain ratio** (C4.5; penalizes many-way splits; CART instead uses binary splits only):

$$\text{Gain ratio} = \frac{\Delta_{\text{info}}}{\text{Split Info}},$$

$$\text{Split Info} = -\sum_{i=1}^{k} \frac{N(v_i)}{N} \log_2 \frac{N(v_i)}{N}$$

- **Continuous split procedure:** (1) sort by attribute; (2) candidate τ = midpoints between consecutive values; (3) compute weighted child impurity for each τ; (4) pick τ minimizing it.
- **Bias/variance:** shallow tree = high bias (underfit); deep tree = high variance (overfit). Total error is minimized at a balance; stopping criteria/pruning trade one for the other.
- **Ensembles** help when base classifiers are independent, high-variance, low-bias.
- **Random forest:** (1) draw random sub-samples of the training data; (2) grow a full tree on each; (3) predict by majority vote. Averaging many deep trees lowers variance while keeping bias low.

09 · k-NN & Naive Bayes

- **Parametric** (fixed # params: linear/logistic regression, Naive Bayes) vs. **nonparametric** (params grow with data: k-NN, trees, forests).
- **k-NN:** training = store the data. Predict $\mathbf{x}$: (1) compute distance to every training point; (2) take the k nearest; (3) hard: majority label (ties possible); soft: $p(c \mid \mathbf{x}, k) = \#\{\text{neighbors with label } c\}/k$.
- k is the hyperparameter controlling complexity: small k = jagged boundary (overfit), large k = smooth (underfit). Choose k on held-out data.
- **Challenges:** prediction cost linear in training size; Euclidean distance ⇒ scale features; curse of dimensionality.
- **Curse of dimensionality:** volume grows exponentially in d (10 bins/dim ⇒ 10^d cells), so data become sparse; to be close, points must be close in *every* dimension; ratio $d_{\max}/d_{\min}$ of pairwise distances $\to 1$ as $d \uparrow$, so "nearest" is meaningless. Unit ball $V_d(R) = \frac{\pi^{d/2}}{\Gamma(d/2+1)} R^d$: its share of the enclosing cube $\to 0$, and nearly all its volume lies in a thin outer shell. Remedies: other metrics (e.g. cosine), reduce dimension.
- **Evaluation:** accuracy = fraction correct (misleading under class imbalance). Confusion matrix (TP, FP, FN, TN); Recall = $\frac{TP}{TP+FN}$; Precision = $\frac{TP}{TP+FP}$.
- **Model-selection protocol:** (1) random train/*validation* split (a final test set stays locked away); (2) train; (3) score on validation; (4) record; (5) repeat many splits and average — for each hyperparameter value; pick the best validation score; report the chosen model's error on the untouched test set.
- **Bayes' rule:** $P(A \mid B) = \frac{P(B|A)\,P(A)}{P(B)}$; prior $P(A)$, posterior $P(A \mid B)$.
- **Probabilistic classifier (MAP):** $P(C_i \mid \mathbf{x}) \propto P(\mathbf{x} \mid C_i)P(C_i)$, so $\hat{C} = \arg\max_{C_i} P(x_1, \ldots, x_d \mid C_i)P(C_i)$.
- **Naive Bayes** assumes conditional independence given the class: $P(x_1, \ldots, x_d \mid C_i) = \prod_j P(x_j \mid C_i)$. Train: estimate $P(C_i)$ and each $P(x_j \mid C_i)$ by counting (or a Gaussian per class for continuous x_j). Predict: $\arg\max_i P(C_i) \prod_j P(x_j \mid C_i)$. A zero count kills the product ⇒ Laplace smoothing (add a small count to every bin).

10 · Linear Regression

- **Model:** $\mathbf{y} = X\beta + \epsilon$; X ($n \times p$) design matrix (column of 1s for intercept), ϵ residual vector. "Linear" = linear in the parameters: $y = \sum_j \beta_j f_j(\mathbf{x})$ for known f_j (polynomial, trend surface, ... all count).
- **Least squares:** $\hat{\beta} = \arg\min_\beta \|X\beta - \mathbf{y}\|_2^2 = \sum_i (\hat{y}_i - y_i)^2$.
- **Normal equations:** $X^T X\beta = X^T\mathbf{y}$ (residual $\perp$ col. space of X); always solvable; unique iff X^TX invertible $\Leftrightarrow$

$\operatorname{rank} X = p$ (needs $n \geq p$, independent columns): $\hat{\beta} = (X^T X)^{-1} X^T \mathbf{y}$.

- With an intercept: $\sum_i (y_i - \hat{y}_i) = 0$, $\bar{\hat{y}} = \bar{y}$, line passes through $(\bar{x}, \bar{y})$.
- **Fit:** $\hat{y}_i = \mathbf{x}_i^T \hat{\beta}$; $\text{TSS} = \sum (y_i - \bar{y})^2$, $\text{RSS} = \sum (y_i - \hat{y}_i)^2$, $\text{ESS} = \sum (\hat{y}_i - \bar{y})^2$; TSS = RSS + ESS (with intercept).

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}} \in [0, 1]$$

- R^2 (coefficient of determination) = fraction of variance of $\mathbf{y}$ explained; uncentered $R^2 = 1 - \text{RSS}/\sum y_i^2$; without intercept the decomposition fails and R^2 can be negative. Not Pearson's r. Adding a column never raises training RSS (training R^2 never falls) — judge on held-out data.
- **Coefficient uncertainty:** $\hat{\sigma}^2 = \text{RSS}/(n-p)$; $\text{Var}(\hat{\beta}) = \hat{\sigma}^2 (X^T X)^{-1}$; $\text{SE}(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 [(X^T X)^{-1}]_{jj}}$; 95% CI $\hat{\beta}_j \pm t_{0.025,\, n-p} \, \text{SE}(\hat{\beta}_j)$; CI containing 0 $\Rightarrow$ not significant. Correlated features / few points inflate $(X^T X)^{-1}$.
- If $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$, least squares = MLE (closed form).

11 · Logistic Regression & Regularization

- **Odds** $= p/(1-p) \in [0, \infty)$; **logit** (log-odds) $= \log \frac{p}{1-p} \in (-\infty, \infty)$; inverse logit = **logistic/sigmoid** $\sigma(z) = \frac{e^z}{1+e^z} = \frac{1}{1+e^{-z}}$.
- **Model:** $\text{logit}\, p(y{=}1 \mid \mathbf{x}) = \beta_0 + \beta_1 x_1 + \cdots$, i.e. $p(y{=}1 \mid \mathbf{x}) = \frac{e^{\beta_0 + \beta^T \mathbf{x}}}{1 + e^{\beta_0 + \beta^T \mathbf{x}}}$. Larger β_1 = steeper sigmoid; $\sigma(\beta_0)$ = predicted probability at $\mathbf{x} = \mathbf{0}$ (equals the base rate only if features are centered); one unit of x_1 multiplies odds by e^{β_1}.
- **Likelihood** (0/1 labels): $P(y_i \mid x_i) = p_i^{y_i}(1-p_i)^{1-y_i}$, $L(\beta) = \prod_i p_i^{y_i}(1-p_i)^{1-y_i}$; log-likelihood $\ell(\beta) = \sum_i [y_i \log p_i + (1-y_i)\log(1-p_i)]$ ($-\ell$ = log-loss); $\sigma'(z) = \sigma(z)(1-\sigma(z))$ gives $\nabla \ell = X^T(\mathbf{y} - \mathbf{p})$. No closed form — gradient ascent: $\beta^{(t+1)} = \beta^{(t)} + \eta \nabla \ell(\beta^{(t)})$ (η = learning rate) until $\nabla \ell \approx 0$.
- **As classifier:** predict 1 if $p(y{=}1 \mid \mathbf{x}) > \tau$ (default $\tau = 0.5$). TP/FP/FN/TN; Precision $= \frac{TP}{TP+FP}$, Recall $= \frac{TP}{TP+FN}$, $F_1 = 2\frac{PR}{P+R}$; lower τ raises recall, lowers precision — tune τ by F_1. ROC = TPR (= recall) vs FPR $= \frac{FP}{FP+TN}$ over τ; AUC = area (overall quality).
- **Overfitting** = high train accuracy, stagnant validation accuracy. **Regularization:** penalize coefficient size:

$$\min_{\beta} \|X\beta - \mathbf{y}\|_2^2 + c\, R(\beta), \quad c \geq 0$$

- **Multicollinearity:** predictors nearly linearly dependent $\Rightarrow$ $X^T X$ (nearly) singular, $\hat{\beta}$ non-unique / huge in magnitude. **Condition number** $\kappa(X) = \sigma_{\max}/\sigma_{\min}$; $\kappa(X^T X) = \kappa(X)^2$ (never form $X^T X$ numerically). Sources: redundant/derived variables, dummy-variable trap, too few observations.
- **Ridge (L2, Tikhonov):** $R = \|\beta\|_2^2$; $\hat{\beta} = (X^T X + cI)^{-1} X^T \mathbf{y}$ — always exists, unique. $c = 0$: OLS; $c \to \infty$: $\hat{\beta} \to 0$; larger c shrinks norm. Shrinks but never zeroes coefficients (no variable selection).
- **LASSO (L1):** $R = \|\beta\|_1$; sets some coefficients exactly to 0 (the ℓ_1 ball has corners on the axes; model selection, interpretability); more zeros as c grows; prediction comparable to ridge. No closed form.
- **Standardize first:** center $\mathbf{y}$ and columns of X (unit variance) so the penalty treats predictors fairly and no intercept is needed. Choose c by cross-validation on held-out data.
- Exhaustive subset selection $= 2^n - 1$ models (infeasible); LASSO does it implicitly.

12 · SVD & Low-Rank Approximation

- **Rank** = # linearly independent rows/columns $\leq \min(m, n)$; full rank vs. rank-deficient. Rank $k \Rightarrow A = CR$, $C \in \mathbb{R}^{m \times k}$, $R \in \mathbb{R}^{k \times n}$: $k(m+n)$ numbers instead of mn.
- **SVD** ($A \in \mathbb{R}^{m \times n}$, $m > n$): $A = U \Sigma V^T$; U ($m \times n$) orthonormal columns = left singular vectors; Σ ($n \times n$) diagonal, $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k > 0 = \sigma_{k+1} = \cdots$; V ($n \times n$) orthogonal, columns = right singular vectors. $A\mathbf{v}_i = \sigma_i \mathbf{u}_i$; $\mathbf{v}_{k+1..n}$ span the null space.
- Always exists; σ_i unique; $\mathbf{u}_i, \mathbf{v}_i$ unique up to sign. $\#\{\sigma_i > 0\} = \operatorname{rank} A$. **By hand:** $A^T A = V \Sigma^2 V^T$, so σ_i^2 = eigenvalues of $A^T A$ (symmetric $\Rightarrow$ real, ≥ 0), $\mathbf{v}_i$ = its orthonormal eigenvectors, $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$.
- **Outer-product form:** $A = \sum_i \sigma_i \mathbf{u}_i \mathbf{v}_i^T$ — sum of rank-1 matrices weighted by σ_i.
- **Frobenius norm:** $\|A\|_F = \sqrt{\sum_{ij} a_{ij}^2} = \sqrt{\sum_i \sigma_i^2}$; $\text{dist}(A, B) = \|A - B\|_F$.
- **Best rank-k approximation** (Eckart–Young): $A^{(k)} = \arg\min_{\text{rank}\, B = k} \|A - B\|_F = U' \Sigma' V'^T = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^T$ (U': first k columns of U; Σ': top-left $k \times k$; V': first k columns of V).

$$\|A - A^{(k)}\|_F^2 = \sum_{i=k+1}^{n} \sigma_i^2$$

- **Low effective rank:** data matrices are usually full rank but well approximated with $k \ll \min(m, n)$; judge from the singular-value spectrum (fraction of $\sum \sigma_i^2$ captured / sharp drop-off).
- **Interpretations:** columns of U' = strongest common patterns (each column of $A^{(k)}$ is a combination of them); entries of $A^{(k)}$ = inner product of a row of $U'\Sigma'$ with a row of V' = latent factors (users $\times$ items).